

Linear equations



1

a i $y = 2x - 7$ ii Yes

b i $y = \frac{x}{2} + 8$ ii Yes

c i $y = \frac{-1}{x} + 6$ ii No

d i $y + \frac{x}{16} = 14$ ii Yes

Tables of values

2

- a
- i The y -value increases by 6.
 - ii The y -value is equal to six times the x -value.
 - iii $y = 6x$

- b
- i The y -value increases by 1.
 - ii The y -value is equal to the x -value plus seven.
 - iii $y = x + 7$

- c
- i The y -value decreases by 1.
 - ii The y -value is equal to the x -value multiplied by -1 and then minus six.
 - iii $y = -x - 6$

- d
- i The y -value decreases by 4.
 - ii The y -value is equal to negative four times the x -value plus two.
 - iii $y = -4x + 2$

- e
- i The y -value increases by 2.
 - ii The y -value is equal to two times the x -value plus five.
 - iii $y = 2x + 5$

3



- a Yes
- b -7 units
- c Because there is a consistent change in y for every one unit change in x .
- d $y = -7x - 5$
- e The change in y -value is the number that goes in front of the x in the rule.

4

- a Yes
- b Yes
- c No
- d Yes
- e Yes
- f No
- g Yes
- h Yes
- i No
- j Yes
- k Yes
- l No
- m Yes
- n Yes
- o No
- p No
- q No
- r Yes

5

- a
 - i 0
 - ii $y = 4x$

iii

x	1	2	3	4	5	14
y	4	8	12	16	20	56

- b
 - i -3
 - ii $y = 2x - 3$

iii

x	1	2	3	4	5	30
y	-1	1	3	5	7	57

- c
 - i 0
 - ii $y = -2x$

iii

x	-16	1	2	3	4	5
y	32	-2	-4	-6	-8	-10

- d
 - i 60
 - ii $y = -8x + 60$

iii

x	-30	1	2	3	4	5
y	300	52	44	36	28	20

e

i -6



ii $y = -7x - 6$

iii

x	1	2	3	4	5	30
y	-13	-20	-27	-34	-41	-216

6

a

i $m = 2, c = 0$

ii $y = 2x$

iii

x	0	1	2	3	4	5	21
y	0	2	4	6	8	10	42

b

i $m = 5, c = 8$

ii $y = 5x + 8$

iii

x	0	1	2	3	4	5	29
y	8	13	18	23	28	33	153

c

i $m = 5, c = -21$

ii $y = 5x - 21$

iii

x	0	1	2	3	4	5	21
y	-21	-16	-11	-6	-1	4	84

d

i $m = -3, c = 24$

ii $y = -3x + 24$

iii

x	0	1	2	3	4	5	65
y	24	21	18	15	12	9	-171

7

a $y = -3x + 2$

b $y = 3x + 4$

c $y = 5x + 4$

d $y = -5x + 3$



Applications

8

a 135 mg

b $t = 70$

9

a 1.4 metres/minute

b $y = 1.4x$

c The change in depth per minute.

d 8.4 metres

e $x = 9$ minutes

10

a 85

b 49 beats per minute

c 3

d $y = 3x + 49$

e The change of Mae's heartbeat in one minute.

f 109 beats per minute